

[RI/2025/Q3]

- 3 The school has been experiencing an increasing number of lost items. You are tasked to create a **Lost and Found** web application and store the data about lost items in a MongoDB database. Your app will allow the Student Affairs staff to submit a new lost item report, view existing lost items in the database, and perform a search on lost items database.

Task 3.1

Write a program to:

- create a Mongo client to connect to database `lostfoundDB`
- use the data in `lost_items.json` to populate the collection `reports`

Save your program code as

`Task3_1_<name>_<index number>.py`

in a folder named `Task3_1_<name>_<index number>`

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Task 3.2

Write a Python program and the necessary files to create a web application. When either the route `"/reports"` or `"/"` is visited, the web application will return a HTML page to display a report table of all the lost items found in the database, under appropriate table headings, sorting the records by date in ascending order.

Save your program code as

`Task3_2_<name>_<index number>.py`

with any additional files / subfolders as needed in a folder named

`Task3_2_<name>_<index number>`

Run the web application and save the returned reports page as

`Task3_2_<name>_<index number>.html`

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Task 3.3

Implement the following functionality to allow for new lost items to be reported.

Student Affairs staff will visit the `"/submit"` route to save information of new lost item to the database.

Create a "Submit Lost Item" HTML form that allows the staff to key in details like `item_name`, `description`, `category`, `date_lost` and `location`.

You should do validation of the `date_lost`, and ensure that invalid dates (for example, `'2025-02-31'` and `'2025/12/12'`) are not accepted, and notify the user that the entered date is invalid.

After submitting the completed form, a new document should be created in the `reports` collection, and the user should be redirected to the `/reports` URL, which will show the new lost item reported, along with existing lost items.

Save your program code as

`Task3_3_<name>_<index number>.py`

with any additional files / subfolders as needed in a folder named

`Task3_3_<name>_<index number>`

Run the web application, add a new report as below, and save the resultant report page as

`Task3_3_<name>_<index number>.html`

- Item Name: Spectacles
- Description: Pink spectacles left outside the Hall
- Category: Accessories
- Date Lost: 2025-07-31
- Location: Hall

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Task 3.4

Implement the following functionality to allow the staff to search for lost items

- by category, or
- by date, or
- by both category and date.

Student Affairs staff will visit the `"/search"` route to perform a search on lost items in the database.

Create a "Search Lost Item" HTML form that contains two form inputs, "Category" and "Date Lost". If the staff to key in only one of the two fields, the search will only show items that fulfil the search field entered. If both fields are entered, only items that fulfil both criteria will be shown in a html table below the search form. Save your program code as

`Task3_4_<name>_<index number>.py`

with any additional files / subfolders as needed in a folder named

`Task3_4_<name>_<index number>`

Search for lost items on date: '2025-07-31' and save output as

`Task3_4_1_<name>_<index number>.html`

Search for lost items with

- date: '2025-07-31' and
- category: "Accessories"

and save the output as

`Task3_4_2_<name>_<index number>.html`

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